

PROJECT FRAMING

Retina RASTA project keypoints

A multimodal retinal OCTA classifier for four RASTA cohorts

The project combines retinal vascular imagery, clinical context, generated biomarkers, and weak-label morphology masks to classify folder-derived RASTA cohorts. MRCC is the clearest CAD-related group in this dataset, but the task is not a direct binary CAD diagnostic endpoint.

Framing and dataset

Cohort	Working interpretation	Patients	Eyes
RETINORM	reference / normal retinal cohort	75	136
ORNET	other retinal / vascular cohort	77	150
FAMILIPO	familial / related cohort	81	156
MRCC	most CAD-related cohort in this framing	32	34

Dataset snapshot

265

patients

476

eyes

211

bilateral

35 / 19

OD-only / OS-only

No patient appears in multiple cohorts. Sup, Deep, and CC are present for all eyes. BMI is missing in 44 rows and dyslipidemia in 2 rows; missing values are handled with learned tokens rather than full-dataset imputation.

The document distinguishes measured evidence from what the available artifacts cannot establish. A cohort label should not be read as a clinical diagnosis.

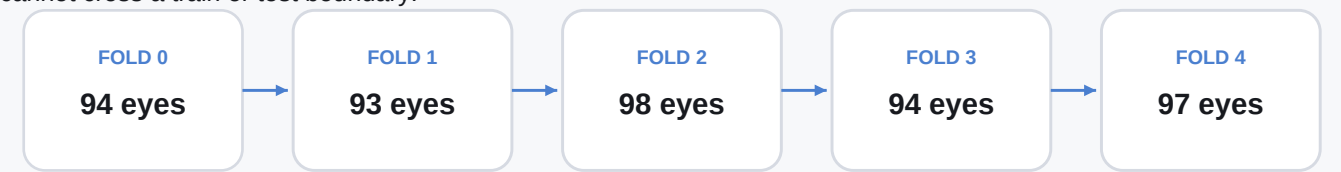
DATA DISCIPLINE

Leakage-safe preprocessing

Patient separation is enforced before fold statistics, validation, calibration, or held-out scoring.

Outer-fold structure

The patient is the split unit and the eye is the prediction unit. OD and OS remain together so both eyes from one person cannot cross a train or test boundary.



476 eyes, zero patient overlap

Controls in the pipeline

Control	Implementation
Split design	Fixed five patient-grouped folds; outer sizes 94 / 93 / 98 / 94 / 97 eyes; zero patient overlap.
Image preparation	Percentile normalization, CLAHE, resize to 224 x 224, and fold-fitted statistics.
Clinical and biomarkers	Training-fold-only normalization, learned missing tokens, and generated biomarkers through the tabular branch.
Augmentation	Synchronized train-only geometric changes across Sup / Deep / CC; image perturbations and clinical masking stay out of validation and test.

Why it matters

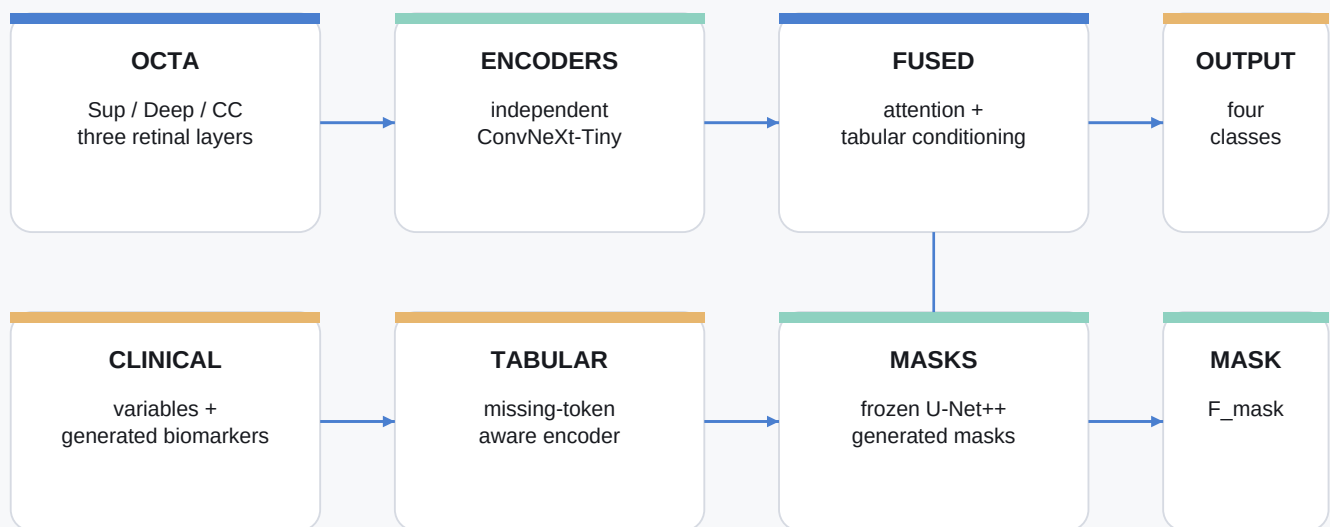
The evaluation claim is anchored to patient identity separation, fold-local normalization, inner validation, calibration, and out-of-fold predictions. This makes the held-out eye score interpretable as a test of a model that did not train on that patient.

MODEL STRUCTURE

Complementary evidence streams

Each modality is prepared independently, then combined through explicit compatibility and prototype operations.

Multimodal four-class model



Clinical reading of the branches

- OCTA supplies layer-specific vascular patterns.
- Clinical variables and biomarkers provide context that an image alone cannot encode.
- The mask branch focuses representation on vessel and flow-deficit structure, but its supervision is weak-label refinement.
- The output is a calibrated four-class probability, not a standalone clinical diagnosis.

Compatibility fusion

The visual and tabular streams form $F_{\text{cond}} = \text{LayerNorm}(F_{\text{octa}} + \text{Proj}_{\text{tab}}(F_{\text{tabular}}))$. The model then concatenates F_{cond} with F_{mask} and projects the result to H in $\mathbb{R}^{512}$.

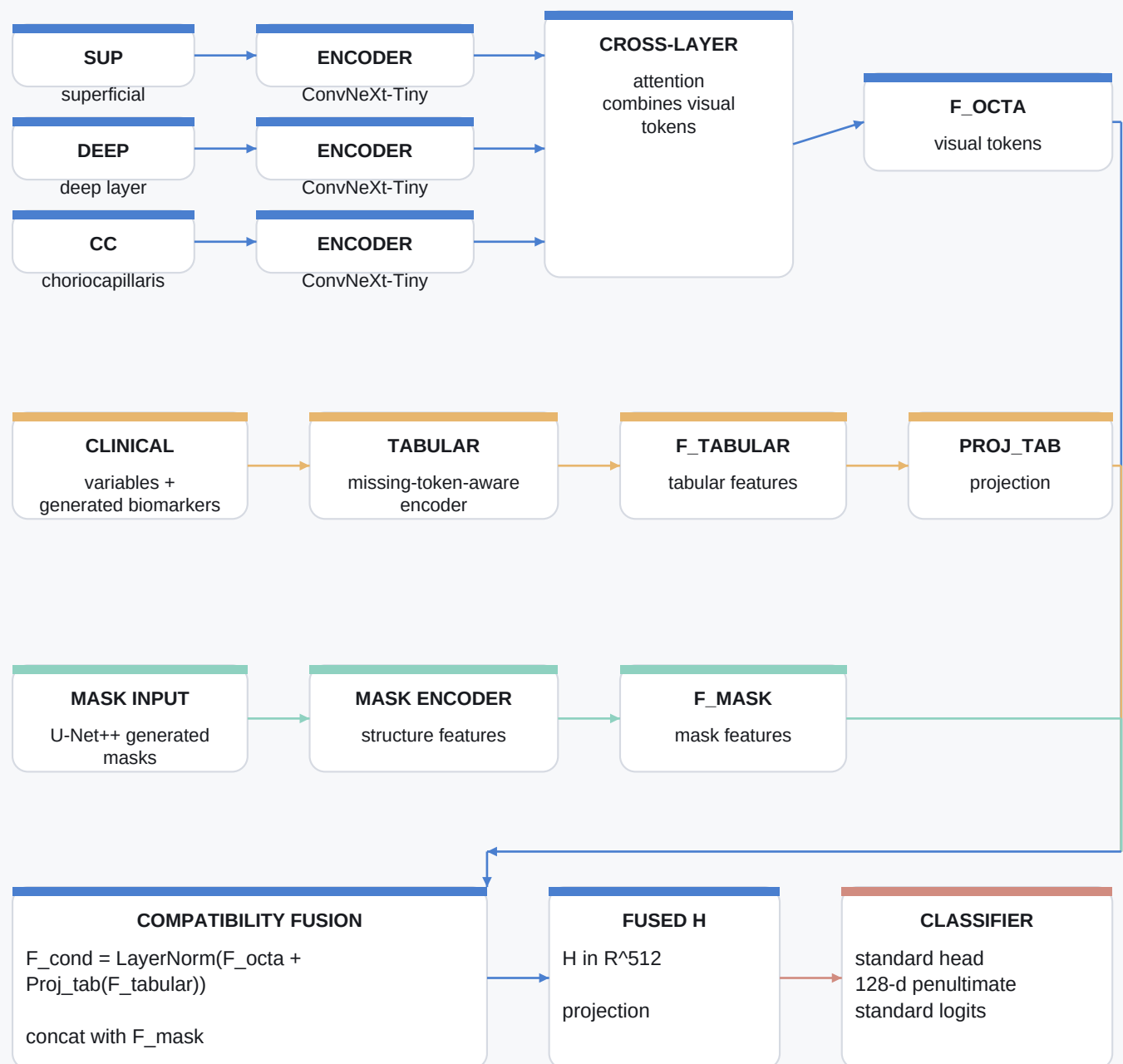
The final classifier includes the standard head, a 128-dimensional penultimate representation, standard class logits, a prototype branch, and temperature scaling.

ARCHITECTURE

How the modalities become one decision

The implemented path preserves each evidence stream until compatibility fusion, then combines standard and prototype logits before calibration.

End-to-end architecture



Prototype logits come from distances to learned class prototypes. The standard and prototype logits are combined with the learned non-negative prototype weight, then temperature scaling produces calibrated four-class probabilities.

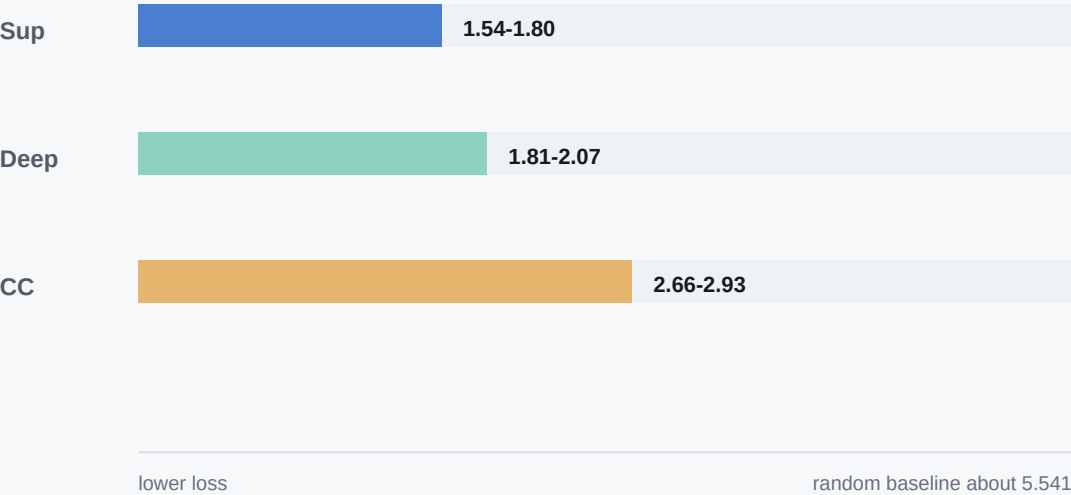
LEARNED REPRESENTATIONS

SimCLR and mask evidence

Self-supervised retinal representations and weak-label morphology features enter the classifier through separate controls.

SimCLR runs

The repository records 15 SimCLR runs at 100 epochs for each retinal layer. Best losses are below the random baseline of about 5.541.



Mask branch

Phase 8 begins with deterministic classical pseudo-masks and trains a compact U-Net++ student against QC-passed pseudo-labels. The QC pass rate is 0.9937. The masks refine representation and generate biomarkers; they are not human annotations.

Student evidence	Dice	IoU
Sup / Deep vessel masks	Dice 0.890-0.898	IoU 0.802-0.815
CC flow-deficit masks	Dice 0.757-0.788	IoU 0.609-0.651

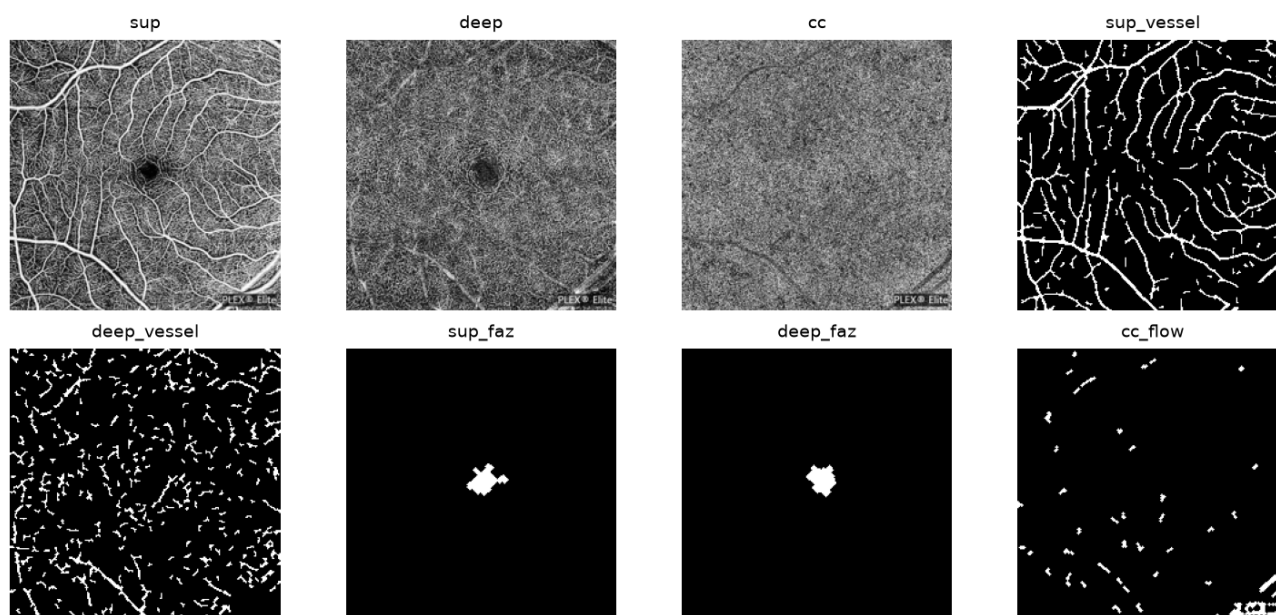
Scores are representative ranges across outer folds. Their value is structural refinement, not clinical ground truth.

MASK BRANCH

Generated morphology masks

One eye, shown across the three retinal slabs and their generated mask outputs.

3CHCA66 OD | label=2



Representative montage: 3CHCA66_OD

The top row shows Sup, Deep, and CC OCTA slabs; the lower row shows classical_improved pseudo-mask outputs for Sup vessel, Deep vessel, Sup FAZ, Deep FAZ, and CC flow. These masks are produced after QC and show the morphology signal used to build mask-branch inputs.

Downstream refinement trains a U-Net++ student against QC-passed pseudo-labels; the masks are not human annotations.

EVALUATION

Classification and calibration

Outer-fold out-of-fold predictions provide the primary performance evidence.

Point estimates from calibrated OOF predictions

0.800

patient accuracy

0.898

patient macro ROC-AUC

0.765

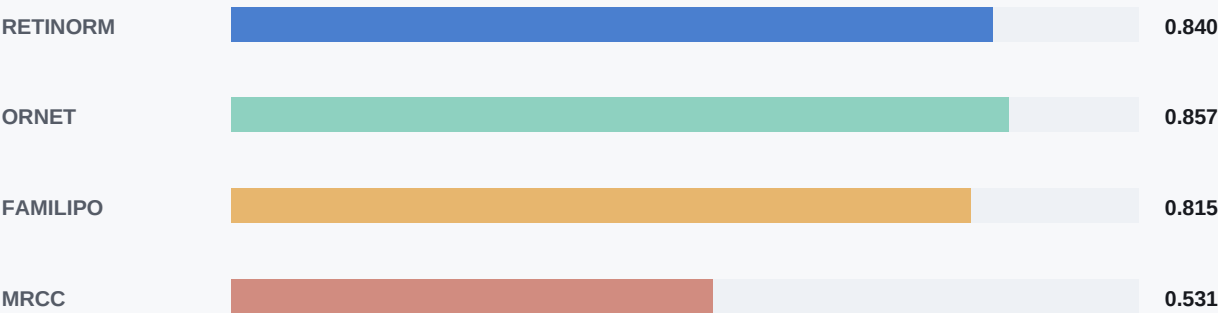
patient macro F1

0.761

balanced accuracy

Level	N	Acc.	95% CI	ROC-AUC	Macro F1	Balanced	Top-2
Patient	265	0.800	0.755-0.842	0.898	0.765	0.761	0.943
Eye	476	0.748	0.705-0.791	0.877	0.699	0.700	0.908

Patient-level recall by cohort



MRCC is the smallest cohort and the limiting class for interpretation.

Calibration

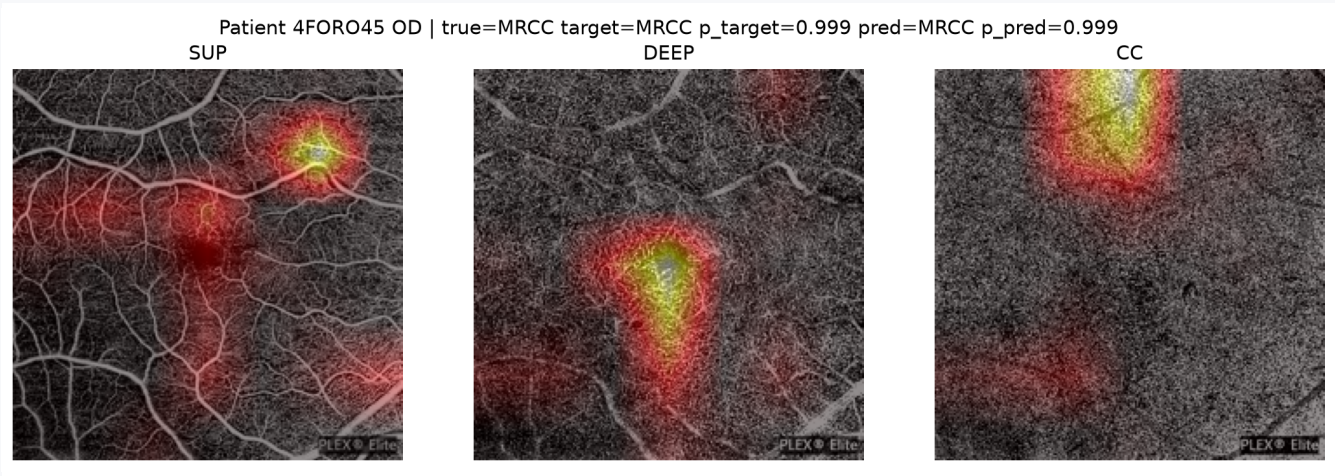
Patient probabilities are formed by averaging calibrated OD and OS probabilities when both eyes are available. Confidence intervals use a 1,000-sample stratified patient-cluster bootstrap. Eye-level ECE is 0.0346; patient-level ECE is 0.0682.

LOCAL EXPLANATION

A correctly predicted MRCC eye

The example connects the held-out prediction to image regions and retinal-layer attention.

4FORO45_OD



true=MRCC, predicted=MRCC | calibrated MRCC probability 0.9990770816802979 (displayed as 99.9%)

Attention row for this eye

Sup	Deep	CC
32.4%	26.3%	41.3%

Layer shares: Sup 32.4%, Deep 26.3%, CC 41.3%.

Grad-CAM localizes image regions supporting the MRCC score in Sup, Deep, and CC. CC has the largest cross-layer attention share for this eye. These percentages describe retinal-layer attention within F_octa, not total fused-modality contributions.

The repository currently has no patient-specific tabular attribution or mask-branch attribution artifact for 4FORO45_OD. The example therefore explains image evidence, retinal-layer weighting, and the mathematical fusion path, but cannot quantify which clinical variable, generated biomarker, or mask feature drove this individual prediction. Grad-CAM and attention are associative explanations, not causal proof.

SCIENTIFIC READING

Strengths and limits

The evidence is strong where the protocol is explicit and bounded where artifacts are incomplete.

Strengths

- Patient-grouped outer folds with inner validation.
- Fold-specific self-supervision and normalization.
- Multimodal design with a frozen mask branch.
- Calibrated out-of-fold evaluation and clustered bootstrap intervals.
- Explicit checkpoint, resume, and status controls.
- Clear separation between image, tabular, and mask evidence.

Limits

- MRCC is small and imbalanced.
- Folder cohorts are not direct CAD labels.
- One dataset is available and there is no external validation.
- Pseudo-masks are weakly supervised, not human ground truth.
- Local explainability is incomplete beyond the available fold 0 artifacts.
- The 4FORO45_OD example has no patient-specific tabular or mask attribution.

Evidence boundary

The reviewed repository artifacts include the dataset audit, SimCLR histories, mask and student metrics, cross-validation summaries, local Phase 13 fold 0 explainability outputs, and the Phase 14 evaluation bundle. Phase 14 evaluation is complete. Local Phase 13 outputs include Grad-CAM and attention artifacts for fold 0; patient-specific tabular and mask attribution artifacts are not present for 4FORO45_OD.

The available evidence supports a careful four-cohort model result with calibrated probabilities and an explicit fusion path. It does not establish a clinically validated binary endpoint or causal feature attribution.

REFERENCE TERMS

Compact glossary

A shared vocabulary keeps the dataset, architecture, and evaluation claims aligned.

Imaging and labels

Term	Meaning
OCTA	Optical coherence tomography angiography, used here as retinal vascular imagery.
Sup / Deep / CC	Superficial, deep, and choriocapillaris retinal layers.
OD / OS	Right eye / left eye.
MRCC	The smallest RASTA cohort and the clearest CAD-related group in this framing.

Evaluation language

Term	Meaning
OOF	Out-of-fold prediction: every sample is scored by a model that did not train on its patient.
ECE	Expected calibration error, a summary of confidence versus observed correctness.
Prototype logits	Logits computed from distances to learned class prototypes.
Temperature scaling	A post-hoc transformation that converts final logits into calibrated probabilities.

Representations

Term	Meaning
F_octa	Visual token representation after cross-layer attention.
F_tabular	Representation from clinical variables and generated biomarkers.
F_mask	Representation from U-Net++ generated masks.
F_cond	LayerNorm compatibility fusion of F_octa and Proj_tab(F_tabular).
H	The 512-dimensional fused representation sent to the standard classifier head.

One example makes the boundary concrete: for 4FORO45_OD, the repository contains image Grad-CAM and cross-layer attention artifacts, but no patient-specific tabular or mask-branch attribution artifact.

FINAL CALCULATION

How the fused model reaches MRCC

The class conclusion is the result of a sequence of learned representations, logits, and calibration.

From three retinal layers to one calibrated class

1

Independent visual encoding

Sup, Deep, and CC images pass through independent ConvNeXt-Tiny encoders.

2

Cross-layer attention

Attention combines visual tokens into F_{octa} . For 4FORO45_OD, the reported layer shares are Sup 32.4%, Deep 26.3%, and CC 41.3%.

3

Tabular and mask branches

Clinical variables and generated biomarkers pass through the missing-token-aware tabular encoder into F_{tabular} . U-Net++ generated masks pass through the mask encoder into F_{mask} .

4

Compatibility fusion

$F_{\text{cond}} = \text{LayerNorm}(F_{\text{octa}} + \text{Proj}_{\text{tab}}(F_{\text{tabular}}))$. The model concatenates F_{cond} with F_{mask} and projects to H in R^{512} .

5

Standard and prototype logits

The standard head maps H to a 128-dimensional penultimate representation and standard class logits. Prototype logits come from distances to learned class prototypes and are combined using the learned non-negative prototype weight.

6

Calibration and decision

Temperature scaling converts final logits into calibrated four-class probabilities. The largest calibrated probability determines the class: MRCC for 4FORO45_OD at 99.9%.